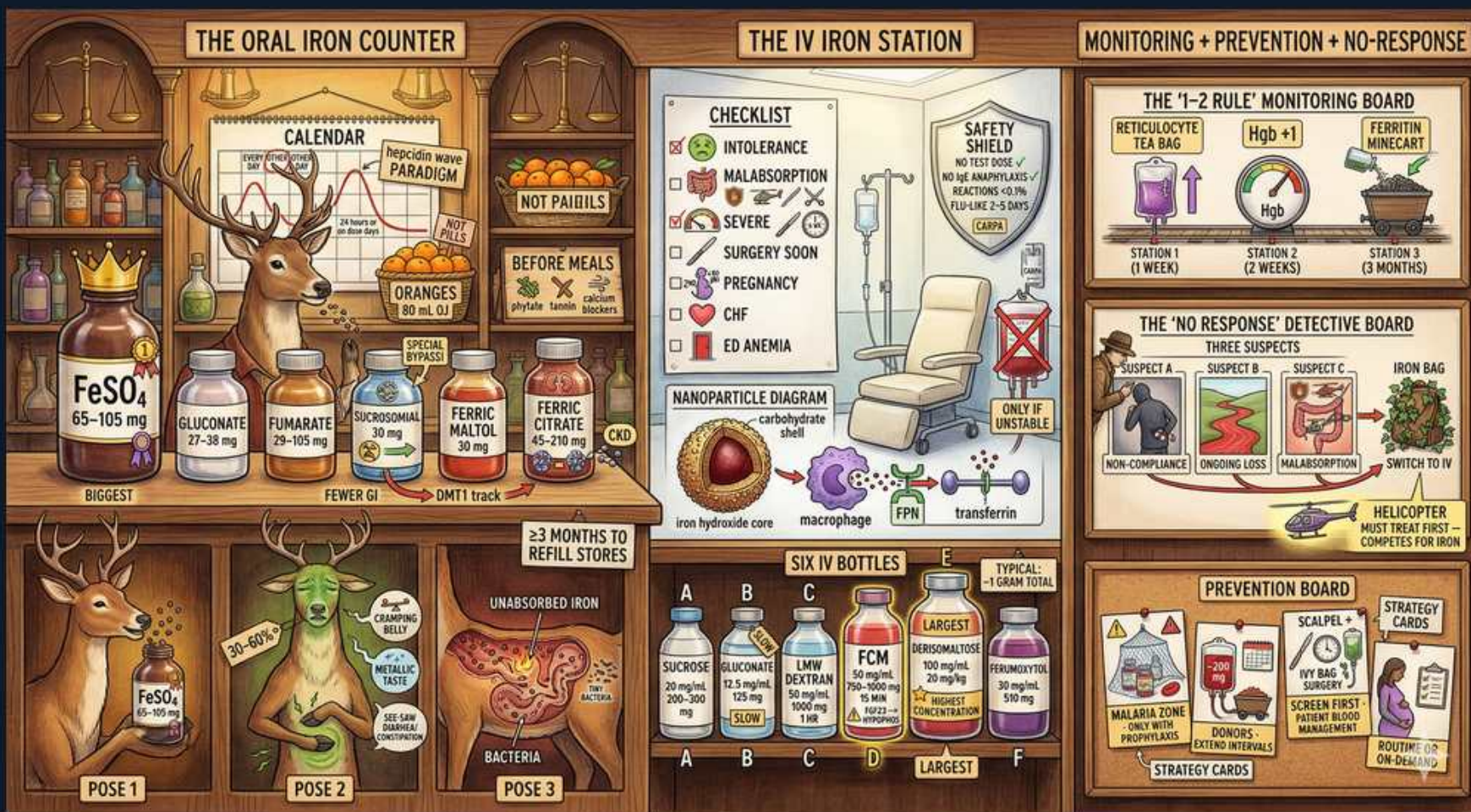


Iron / Microcytic — iron deficiency treatment (oral, IV, monitoring)



1. FeSO4 — first-line oral

WHAT YOU SEE

Big crowned 'FeSO₄' 65–105 mg' bottle, 'BIGGEST'

WHAT IT ENCODES

Ferrous SULFATE is first-line oral therapy: a 325 mg tablet provides ~65 mg ELEMENTAL iron (ferrous sulfate is ~20% elemental). It is the cheapest, most studied option; total daily target is ~100–200 mg elemental iron, increasingly given as a single daily or alternate-day dose.

BOARD TRAP

WRONG: starting an expensive ferric salt or IV iron first-line for simple, uncomplicated IDA. Ferrous sulfate is the proven, low-cost standard; reserve newer ferric formulations and IV iron for intolerance, malabsorption, or failure of oral therapy.

2. Take before meals

WHAT YOU SEE

'BEFORE MEALS', oranges (vit C), empty-stomach deer

WHAT IT ENCODES

Take oral iron on an EMPTY stomach (1 hr before or 2 hr after meals), ideally with vitamin C / orange juice to enhance Fe³⁺→Fe²⁺ reduction and absorption. Separate it from calcium/dairy, antacids, PPIs, and tea/coffee, which all impair uptake.

BOARD TRAP

WRONG: routinely advising patients to take iron WITH meals or antacids to reduce nausea. Food and acid-suppressants markedly cut absorption; if GI upset is limiting, lower the dose or switch to alternate-day dosing rather than pairing it with food/PPIs that defeat the therapy.

3. Alternate-day dosing

WHAT YOU SEE

Calendar 'PARADIGM', spaced dosing, 'FEWER GI'

WHAT IT ENCODES

Newer evidence: ONCE-daily or ALTERNATE-day dosing achieves greater fractional absorption than divided multiple-daily doses, because each dose triggers a transient hepcidin rise that blunts absorption of subsequent doses for ~24 hr. Spacing doses also reduces GI side effects and improves adherence.

BOARD TRAP

WRONG: thinking three-times-daily dosing absorbs more iron and works faster. The dose-induced hepcidin spike means TID is NOT superior — alternate-day/once-daily dosing actually absorbs more per milligram given and is better tolerated. More frequent ≠ more absorbed.

4. Lower-elemental salts

WHAT YOU SEE

Row: gluconate 27–36, fumarate 33, sucrosomial, ferric maltol, ferric citrate

WHAT IT ENCODES

Alternative oral salts differ in elemental content and tolerability: ferrous GLUCONATE (~27–36 mg elemental/tab, ~12%), ferrous FUMARATE (~33% elemental, ~106 mg/325 mg tab), plus sucrosomial iron, ferric maltol, and ferric citrate (also lowers phosphate in CKD). Lower elemental dose can mean fewer GI symptoms.

BOARD TRAP

WRONG: comparing tablets by total milligrams rather than ELEMENTAL iron. A 325 mg fumarate tablet (~106 mg elemental) is not equivalent to a 325 mg sulfate tablet (~65 mg elemental) — always dose by elemental iron, or you may under- or over-treat.

5. 3 months to refill stores

WHAT YOU SEE

'≥3 MONTHS to refill stores' sign

WHAT IT ENCODES

Continue oral iron for AT LEAST 3 months AFTER the hemoglobin normalizes to rebuild ferritin/storage iron — total course is often 4–6 months. Stopping when Hb corrects leaves stores empty and the anemia relapses. Recheck ferritin/CBC to confirm repletion.

BOARD TRAP

WRONG: stopping iron as soon as the hemoglobin normalizes. Hb corrects before stores refill; discontinuing then leaves ferritin low and guarantees recurrence. Treat to a target ferritin, not just a normal Hb.

6. Hepcidin/DMT1 absorption block

WHAT YOU SEE

Pose-2 deer with 'DMT1 track', single-dose hepcidin rise

WHAT IT ENCODES

Each oral iron dose transiently raises serum hepcidin, which degrades enterocyte ferroportin and blunts absorption of the next dose for roughly 24 hours. This pharmacology is exactly why ALTERNATE-day/once-daily dosing maximizes net absorption per dose taken.

BOARD TRAP

WRONG: assuming absorption is simply proportional to how often you dose. The hepcidin feedback caps short-interval absorption, so doubling frequency does not double uptake — it can lower fractional absorption and add side effects.

7. Unabsorbed iron / GI side effects

WHAT YOU SEE

Pose-3 gut, 'UNABSORBED IRON', bacteria, dark stool

WHAT IT ENCODES

Unabsorbed luminal iron causes the classic GI side effects — constipation, nausea, epigastric pain, metallic taste, and harmless dark/black stools — which are the leading reason for nonadherence. Manage by dose reduction, alternate-day dosing, or switching salts; add a stool softener for constipation.

BOARD TRAP

WRONG: mistaking benign iron-induced black stool for melena (GI bleeding). Iron causes greenish-black, NON-tarry, guaiac-negative stool; true melena from upper GI bleed is sticky, tarry, and occult-blood positive. Don't trigger an unnecessary bleed workup — but also don't ignore real melena attributing it to iron.

8. IV iron indications checklist

WHAT YOU SEE

CENTER checklist: intolerance, malabsorption, severe, surgery soon, pregnancy, CHF, CKD anemia

WHAT IT ENCODES

Use IV iron when: oral iron fails or isn't tolerated; MALABSORPTION (celiac, gastric bypass, IBD); ongoing/severe blood loss outpacing oral repletion; chronic kidney disease (especially on dialysis/ESAs); heart failure (functional iron deficiency); 2nd/3rd-trimester pregnancy; and pre-operative correction when surgery is imminent.

BOARD TRAP

WRONG: defaulting to IV iron for simple uncomplicated IDA with a working gut. Oral iron is first-line; IV is costlier and carries infusion-reaction risk. Reserve IV for the specific indications above — choosing IV when oral would work is a board pitfall.

9. IV safety / anaphylaxis

WHAT YOU SEE

'SAFETY SHIELD', 'ONLY IF UNSTABLE', nanoparticle diagram

WHAT IT ENCODES

Modern IV iron formulations (ferric carboxymaltose, iron sucrose, ferric derisomaltose, ferumoxytol) are SAFE with low reaction rates; have anaphylaxis precautions available. The dangerous high-molecular-weight iron DEXTRAN (which required a test dose) is largely obsolete. Mild self-limited 'Fishbane' flushing reactions don't require stopping permanently.

BOARD TRAP

WRONG: avoiding IV iron entirely out of fear of anaphylaxis, or routinely giving a test dose for modern agents. Serious reactions are rare with current preparations; the test-dose requirement applied to old high-molecular-weight dextran, not to ferric carboxymaltose/iron sucrose.

10. FCM — largest single dose

WHAT YOU SEE

Six IV bottles; 'FCM' marked 'LARGEST'

WHAT IT ENCODES

Ferric carboxymaltose (FCM) and ferric derisomaltose allow large single-visit doses (up to ~1000+ mg), correcting deficits in 1–2 visits. A notable FCM side effect is transient, usually asymptomatic HYPOPHOSPHATEMIA (from increased FGF23); check phosphate with repeated dosing.

BOARD TRAP

WRONG: forgetting hypophosphatemia as the characteristic FCM adverse effect. Repeated or high-dose FCM can cause symptomatic hypophosphatemia (bone pain, osteomalacia, weakness) — monitor phosphate, especially with recurrent infusions, rather than attributing symptoms elsewhere.

11. 1-2 rule reticulocyte response

WHAT YOU SEE

RIGHT 'THE 1-2 RULE' monitoring: retic at 1 wk, Hgb +1 at 2 wk

WHAT IT ENCODES

Monitoring of effective iron therapy: RETICULOCYTOSIS appears at ~5–7 days (~1 week), and HEMOGLOBIN rises ~1 g/dL every 2–3 weeks, with full correction over ~6–8 weeks. A reticulocyte bump confirms the marrow is responding and the patient is adherent.

BOARD TRAP

WRONG: declaring oral iron a failure after only a few days because Hb hasn't moved. Hb rises slowly (~1 g/dL per 2–3 weeks); the first thing to check is the reticulocyte response at ~1 week. Re-evaluating Hb too early causes premature, unnecessary switches to IV iron.

12. No-response detective board

WHAT YOU SEE

'NO RESPONSE' detective: non-compliance, ongoing loss, malabsorption, wrong dx

WHAT IT ENCODES

When iron therapy fails, work the differential: (1) NONADHERENCE (most common — GI side effects), (2) ongoing blood loss exceeding intake, (3) MALABSORPTION (undiagnosed celiac, gastric bypass, PPI, H. pylori), and (4) WRONG DIAGNOSIS (thalassemia, ACD, sideroblastic — these don't respond to iron).

BOARD TRAP

WRONG: assuming nonresponse automatically means the patient needs IV iron. First confirm adherence, exclude ongoing bleeding and malabsorption (check anti-tTG for celiac), and re-verify the diagnosis — switching a misdiagnosed thalassemia or ACD patient to IV iron just adds iron they don't need.